

4MA1_2H_ch1_numbers_and_number_system

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1. 4MA1/2H_Winter_2024_Q15

Use algebra to show that $0.\dot{7}\dot{6}3 = \frac{42}{55}$

2 marks

2. 4MA1/2H_Winter_2024_Q14

Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n
Show your working clearly.

3 marks

3. 4MA1/2H_Winter_2024_Q9

Nisha invests 20000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.

Give your answer correct to the nearest euro.

3 marks

4. 4MA1/2H_Winter_2024_Q4

N is a number.
17% of N is 357

(a) Work out the value of N

$N = \dots\dots\dots$
(2)

In 2019, the population of a village was 650
In 2020, the population of the village was 806

(b) Work out the percentage increase in the population.

$\dots\dots\dots$
(3)

5. 4MA1/2H_Winter_2024_Q2

The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres

(1)

(b) Write down the lower bound of the length of the table.

..... metres

(1)

6. 4MA1/2H_Winter_2024_Q1

Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

3 marks

7. 4MA1/2HR_Summer_2024_Q19

Aviv goes on a cycle journey.

For the cycle journey

average speed = 19km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.

Give your answer correct to 3 significant figures.

..... km
(3)

8. 4MA1/2HR_Summer_2024_Q18

Use algebra to show that $0.\dot{3}0\dot{6} = \frac{34}{111}$

(2)

9. 4MA1/2HR_Summer_2024_Q9

- (a) Write 6.04×10^5 as an ordinary number.

(1)
- (b) Write 0.000 07 in standard form.

(1)
- (c) Work out $\frac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$
Give your answer in standard form.

(2)

10. 4MA1/2HR_Summer_2024_Q8

In a sale, all normal prices are reduced by 17%
The sale price of a fridge is 6225 rupees.
Work out the normal price of the fridge.

..... rupees

(3)

11. 4MA1/2HR_Summer_2024_Q7

$$\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$$

Find the value of n

(2)

12. 4MA1/2HR_Summer_2024_Q1

Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(3)

13. 4MA1/2H_Summer_2024_Q19

$$G = \frac{c}{2f - 3h}$$

- $c = 8$ correct to the nearest whole number
- $f = 6.62$ correct to 2 decimal places
- $h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G
Give your answer correct to 3 decimal places.
Show your working clearly.

(3)

14. 4MA1/2H_Summer_2024_Q9

Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.
Give your answer correct to the nearest koruna.

..... koruna
(3)

15. 4MA1/2H_Summer_2024_Q8

Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

(5)

16. 4MA1/2H_Summer_2024_Q6

Change a speed of w metres per second to a speed in kilometres per hour.
Give your answer in terms of w in its simplest form.

..... kilometres per hour
(3)

Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

(3)

A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.
The plane flies at an average speed of 820 km/h.
Work out the total distance the plane flies.

..... km
(3)

Here are eight numbers written in order of size

h 6 7 8 j 16 k k

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h , the value of j and the value of k

(3)

A rectangle has length L and width W

L is increased by 20%

W is decreased by 35%

Calculate the percentage reduction in the area of the rectangle.

.....%

3 marks

Teresa invests \$2000 for 3 years in a savings account.
She gets 4% each year compound interest.

- (a) How much money will Teresa have in her savings account at the end of 3 years?
Give your answer correct to the nearest dollar.

\$
(3)

Sam invested \$ T
The value of his investment decreased by 9% each year.

At the end of the first year, the value of Sam's investment was \$1365

- (b) Work out the value of T

.....
(3)

60 students sat a Mathematics exam.

The mean mark for the 32 students in Class A was 55

The mean mark for the 28 students in Class B was 52

Find the mean mark for all 60 students.

.....

3 marks

- (a) Find the highest common factor (HCF) of 200 and 420

.....
(2)

$$A = 2^3 \times 3 \times 5 \times 7^2$$

$$B = 2 \times 3^2 \times 7$$

$$C = 3 \times 5^2 \times 11$$

- (b) Find the lowest common multiple (LCM) of A , B and C
Write your answer as a product of powers of prime factors.

.....
(2)

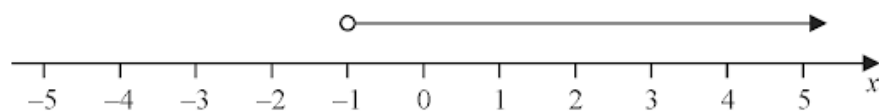
$$-8 < 2y \leq 2$$

y is an integer.

(a) Find all the possible values of y

.....
(2)

(b) Write down the inequality shown on the number line.



.....
(1)

Show that $4\frac{2}{3} \div 1\frac{5}{6} = 2\frac{6}{11}$

3 marks

Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

Bank G 1.6% per year compound interest

Bank H 2.9% interest added after two years

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?

..... Swiss francs

4 marks

The weight of a cake is 2.75 kg, correct to 2 decimal places.

(a) Write down the lower bound of the weight of the cake.

..... kg
(1)

(b) Write down the upper bound of the weight of the cake.

..... kg
(1)

Penny has worked out $\frac{81.3 \times 59.2}{1.9^2}$ on her calculator.

Her answer is 13 332.299 17

Penny's answer is not sensible.

(c) By rounding each number to one significant figure, work out a suitable estimate to show that her answer is not sensible.
Show your working clearly.

(2)

Change a speed of 90 kilometres per hour to a speed in metres per second.
Show your working clearly.

..... m/s

3 marks

Show that $3\frac{5}{7} \div 1\frac{5}{8} = 2\frac{2}{7}$

3 marks

Show that $\frac{2\sqrt{3}}{\sqrt{3}-1}$ can be written in the form $a + \sqrt{a}$ where a is an integer.

Show your working clearly.

3 marks

Feruzi invests 80 000 Kenyan shillings (KES)
He invests the money for 3 years at $x\%$ compound interest each year.
At the end of 3 years, the total interest he receives is 6151.25 KES
Work out the value of x

$x =$

3 marks

Gemara works as a taxi driver.

Last week, he recorded the following information about the distances he drove.

For the 5 days from Monday to Friday, the mean number of kilometres he drove was 104

For the 7 days from Monday to Sunday, the mean number of kilometres he drove was 127

On Saturday, Gemara drove 132 kilometres.

Work out the number of kilometres he drove on Sunday.

..... kilometres

3 marks

In his previous job, Pierre was paid 400 euros in total for working a 5-day week.

In his new job, Pierre is paid 14 euros per hour.

In his new job, Pierre works for 7 hours each day for a 5-day week.

(a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week.

..... %
(4)

Marie changes her job.

Her salary decreases by 6%

Her new salary is 23 030 euros.

(b) Work out Marie's salary before she changes her job.

..... euros
(3)

Anjali travels on the Eurostar train from Paris to Amsterdam.

The distance the train travels between Paris and Amsterdam is 515 km.

The time taken by the train to travel between Paris and Amsterdam is 3 hours 18 minutes.

Work out the average speed of the train.

Give your answer in km/h correct to the nearest whole number.

..... km/h

3 marks

Sandeep wants to buy some packets of pens and some boxes of pencils for his stationery shop.

Each packet of pens contains 9 pens.

Each box of pencils contains 12 pencils.

Each packet of pens costs £7.60

Each box of pencils costs £4.80

Sandeep can only buy full packets of pens and full boxes of pencils.

He wants to buy exactly the same number of pens as pencils.

Work out the minimum amount Sandeep needs to pay.

£.....

4 marks

A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

..... g/cm^3

4 marks

Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17

After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.

After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

.....

4 marks

Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

3 marks

Charlotte buys a painting for \$680

The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.

Give your answer correct to the nearest \$

\$.....

3 marks

(a) Write 76 000 000 in standard form.

.....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

.....
(1)

Nancy has some coins with a total value of 85 pence.

She has only 2 pence coins and 5 pence coins.

The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

.....
4 marks

Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

3 marks

$$a = \sqrt{8} + 4$$

$$b = \sqrt{8} - 4$$

$(a - b)(a + b)$ can be written in the form $y\sqrt{4y}$

Find the value of y

Show your working clearly.

$$y = \dots\dots\dots$$

3 marks

The table gives information about the population and the total amount of money, in dollars, spent on healthcare for two countries in 2016

Country	Total population	Total spent on healthcare (\$)
Austria	8.7×10^6	4.2×10^{10}
Luxembourg	6.3×10^5	3.7×10^9

Work out how much more was spent **per person** on healthcare in Luxembourg than in Austria.

Give your answer correct to the nearest whole number.

..... dollars

3 marks

A scientist is investigating the weight of 50 tigers.

Here is some information about these tigers.

	Type of tiger	
	Siberian	Bengal
Number of tigers	22	28
Mean weight of tigers (kg)	260	

The mean weight of all 50 tigers is 218 kg

Work out the mean weight of the Bengal tigers.

..... kg

3 marks

Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

..... dollars

3 marks

Mary saves for a holiday each year.

In 2020 she saved a total of \$720

In 2021, each month she saved \$78

The total amount Mary saved in 2021 was $P\%$ more than the total she saved in 2020

(a) Work out the value of P

Show that $\frac{\sqrt{12}}{\sqrt{3} + 2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

3 marks

49. 4MA1/2H_Winter_2022_Q12

$$a = 6 \times 10^{40}$$

Work out the value of a^3

Give your answer in standard form.

.....
3 marks

The table gives information about the average house price in England in 2018 and in 2019

Year	2017	2018	2019
Average house price (£)		228 314	231 776

- (a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

..... %
(2)

The average house price in 2019 was 7.7% greater than the average house price in 2017

- (b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£
(3)

Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.

.....
3 marks

A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio

number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

.....

5 marks

53. 4MA1/2H_Winter_2022_Q4

Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

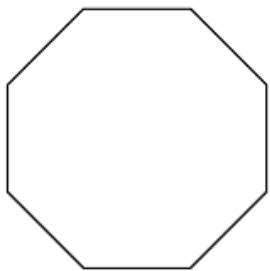


Diagram **NOT**
accurately drawn

(a) Write down the lower bound of the length of each side of the octagon.

..... mm
(1)

(b) Write down the upper bound of the length of each side of the octagon.

..... mm
(1)

Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

3 marks

Kaidan and Sonja went on two different car journeys.

For Kaidan's journey

distance = 80 km correct to the nearest 5 km

time = 2.7 hours correct to 1 decimal place

For Sonja's journey

distance = 33 km correct to 2 significant figures

time = 1 hour correct to the nearest 0.1 hour

Kaidan says,

"My average speed could have been greater than Sonja's average speed."

By considering bounds, show that Kaidan is correct.

Show your working clearly.

4 marks

The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

(a) Write 8.8×10^6 as an ordinary number.

.....
(1)

(b) Which of these cities had the least population in 2018?

.....
(1)

(c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018
Give your answer in standard form correct to 2 significant figures.

.....
(2)

57. 4MA1/2HR_Summer_2022_Q8

A cinema increased the cost of an adult ticket by 12%

After the increase, the cost of an adult ticket was £18.20

Work out the cost of an adult ticket before the increase.

£

3 marks

Write 1200 as a product of powers of its prime factors.
Show your working clearly.

.....

3 marks

$$X = \frac{2a - b}{f}$$

$a = 7.5$ correct to 1 decimal place.

$b = 3.42$ correct to 2 decimal places.

$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X

Show your working clearly.

.....
3 marks

Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W

$W =$

3 marks

Pasha invests 50 000 dollars in a savings account for 4 years.
He gets 1.3% per year compound interest.

Work out how much money Pasha will have in his savings account at the end of 4 years.
Give your answer correct to the nearest dollar.

..... dollars

3 marks

Behnaz makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7 : 5 : 3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

.....
5 marks

Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

3 marks

Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

3 marks

An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

3 marks

(a) Simplify $8^2 \times \sqrt[3]{4^6}$

Give your answer in the form 2^a where a is an integer.
Show each stage of your working clearly.

.....
(3)

Given that $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$ where $n > 0$

(b) find the value of n .

$n =$
(4)

67. 4MA1/2HR_Winter_2021_Q14

Simon bought a house at the beginning of 2018

The value of Simon's house had decreased by 15% by the end of 2018

The house increased in value during both 2019 and 2020

The percentage increases in the value of the house during 2019 and 2020 were the same.

The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018

Calculate the percentage increase in the value of the house during 2019

.....%

(Total for Question is 4 marks)

The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean	Length of coastline (km)
Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Indian	6.653×10^4
Southern	1.797×10^4

(a) Which ocean has the greatest length of coastline?

.....
(1)

(b) Calculate the difference between the length of the Atlantic Ocean’s coastline and the length of the Southern Ocean’s coastline.
Give your answer in standard form.

..... km
(2)

On Wednesday, the price of 1 litre of petrol was £1.26

The price of petrol on Wednesday was 5% more than the price of petrol on the previous Monday.

Calculate the price of 30 litres of petrol on the previous Monday.

£

(Total for Question is 3 marks)

The people working for a company work in Team A or in Team B.

number of people in Team A : number of people in Team B = 3 : 4

$\frac{4}{5}$ of Team A work full time.

24% of Team B work full time.

Work out what fraction of the people working for the company work full time.
Give your fraction in its simplest form.

(Total for Question is 3 marks)

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

Find the lowest common multiple (LCM) of A and B .

Give your answer as a product of powers of prime numbers.

.....
(Total for Question is 2 marks)

There are 5 cocoa pods in a bag.

The mean weight of the 5 cocoa pods is 398 grams.

A sixth cocoa pod is put into the bag.

The mean weight of the 6 cocoa pods is 401 grams.

Work out the weight of the sixth cocoa pod that is put into the bag.

..... grams

(Total for Question is 3 marks)

(a) Write down the value of y^0

.....
(1)

(b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

.....
(3)

Gladys buys a table for \$465 to sell in her shop.

She sells the table for \$520

- (a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

..... %
(3)

Gladys has a sale in her shop.

She decreases all the normal prices by 12%

The normal price of an armchair was \$550

- (b) Work out the sale price of the armchair.

\$.....
(3)
